

Rational Functions & Asymptotes

ANSWER KEY



Vertical asymptotes and holes

- Find the vertical asymptotes and holes of $f(x) = \frac{x^2 - 9}{x^2 - x - 12}$.
Factor: $f(x) = \frac{(x-3)(x+3)}{(x-4)(x+3)}$. The factor $(x+3)$ cancels, giving a hole at $x = -3$. The remaining denominator factor $(x-4)$ gives a vertical asymptote at $x = 4$.
- Find the y-coordinate of the hole in the function from the previous problem.
Simplified form (away from the hole): $f(x) = \frac{x-3}{x-4}$. At $x = -3$: $f(-3) = \frac{-3-3}{-3-4} = \frac{-6}{-7} = \frac{6}{7}$. Hole at $(-3, \frac{6}{7})$.

Horizontal and slant asymptotes

- Find the horizontal asymptote of $f(x) = \frac{3x^2 - 2x + 1}{5x^2 + 4}$, using the degrees of numerator and denominator.
Degrees are equal (both 2), so the horizontal asymptote is the ratio of leading coefficients: $y = \frac{3}{5}$.
- Find the horizontal asymptote of $g(x) = \frac{2x + 7}{x^2 - 1}$.
Degree of numerator (1) is less than degree of denominator (2), so the horizontal asymptote is $y = 0$.
- Explain why $h(x) = \frac{x^2 + 1}{x - 2}$ has a slant (oblique) asymptote instead of a horizontal one, and find it using polynomial division.
The numerator's degree (2) is exactly one more than the denominator's degree (1), producing a slant asymptote. Dividing: $x^2 + 1 = (x - 2)(x + 2) + 5$, so $h(x) = x + 2 + \frac{5}{x - 2}$. As $x \rightarrow \pm \infty$, the remainder term vanishes, leaving the slant asymptote $y = x + 2$.

Visualizing a rational function's asymptotes

- The graph shows $f(x) = \frac{1}{x-2} + 1$, a transformed reciprocal function.
 - Identify the vertical asymptote shown. Vertical asymptote at $x = 2$, where the two branches shoot off toward $\pm \infty$.
 - Identify the horizontal asymptote shown. Horizontal asymptote at $y = 1$, which both branches approach far from $x = 2$ (this is the reciprocal function $\frac{1}{x}$ shifted right 2 and up 1).

End behavior of rational functions

- 7 Describe the end behavior of $f(x) = \frac{2x^3 - x}{x^2 + 1}$ as $x \rightarrow \pm \infty$ (note the numerator's degree exceeds the denominator's by 1, so the function behaves like its slant asymptote).
- Since the degree of the numerator (3) exceeds the denominator (2) by exactly 1, f has a slant asymptote and $f(x) \rightarrow \pm \infty$ as $x \rightarrow \pm \infty$, following the line found by dividing $2x^3 - x$ by $x^2 + 1$, approximately $y = 2x$ for large $|x|$.
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Domain and intercepts

- 8 Find the domain, x-intercept(s), and y-intercept of $f(x) = \frac{x - 5}{x^2 - 4}$.
- Domain: all reals except $x = \pm 2$ (where the denominator is zero), so $(-\infty, -2) \cup (-2, 2) \cup (2, \infty)$. x-intercept: set numerator to 0: $x = 5$, giving $(5, 0)$. y-intercept: $f(0) = \frac{-5}{-4} = \frac{5}{4}$, giving $(0, \frac{5}{4})$.
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Solving rational equations and inequalities

- 9 Solve $\frac{x+1}{x-3} = 2$, checking that the solution doesn't create a zero denominator.
- $x + 1 = 2(x - 3) = 2x - 6$, so $-x = -7$, $x = 7$. Check: $x - 3 = 4 \neq 0$, valid. Solution: $x = 7$.
- 10 Solve the inequality $\frac{x-1}{x+2} > 0$ by determining the sign of the expression on each interval created by the zero and the excluded value.
- Critical points: $x = 1$ (zero) and $x = -2$ (excluded). Testing intervals: for $x < -2$, both factors negative, ratio positive. For $-2 < x < 1$, numerator negative, denominator positive, ratio negative. For $x > 1$, both positive, ratio positive. Solution: $x < -2$ or $x > 1$, i.e. $(-\infty, -2) \cup (1, \infty)$.
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Visualizing a function with a slant asymptote

- 11 The graph shows $f(x) = \frac{x^2}{x-1}$, which has a slant asymptote.
- a** Find the vertical asymptote and confirm it matches the graph. Vertical asymptote at $x = 1$, matching the break in the graph shown there.
- b** Use polynomial division to find the slant asymptote equation. Dividing: $x^2 = (x - 1)(x + 1) + 1$, so $f(x) = x + 1 + \frac{1}{x-1}$. Slant asymptote: $y = x + 1$, matching the line the branches hug for large $|x|$.
- c** Explain why this function has no horizontal asymptote. Since the numerator's degree (2) exceeds the denominator's degree (1), the function grows without bound rather than leveling off, so there is no horizontal asymptote — only the slant asymptote found above.
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Building a rational function from given features

- 12** Construct a rational function with a vertical asymptote at $x = 3$, a hole at $x = -1$, and a horizontal asymptote at $y = 2$.

Start with $\frac{2(x+1)}{x+1} \cdot \frac{1}{x-3}$ -style construction: use $f(x) = \frac{2(x+1)^2}{(x+1)(x-3)}$. The common factor $(x+1)$ creates the hole at $x = -1$; the remaining $(x-3)$ in the denominator gives the vertical asymptote; matching leading coefficients (2 over 1) gives horizontal asymptote $y = 2$.

- 13** Explain why a rational function cannot have both a horizontal asymptote and a slant asymptote at the same time.

A horizontal asymptote occurs when the numerator's degree is less than or equal to the denominator's degree, while a slant asymptote requires the numerator's degree to be exactly one more than the denominator's. These are mutually exclusive degree conditions, so a rational function has at most one of the two (or neither, if the numerator's degree exceeds the denominator's by more than 1).

Rational functions in context

- 14** The concentration of a drug in the bloodstream t hours after a dose is modeled by $C(t) = \frac{50t}{t^2 + 4}$ (mg/L). Find the horizontal asymptote and interpret it in context.

Degree of numerator (1) is less than degree of denominator (2), so the horizontal asymptote is $y = 0$. In context, this means the drug concentration approaches 0 mg/L as time goes on — the drug is eventually fully eliminated from the bloodstream.